

Sujay Seeram

Software Systems, AI Infrastructure & High-Performance Runtimes Engineer

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Location: Coimbatore, Tamil Nadu / Andhra Pradesh, India

ENGINEERING PROFILE & CORE PHILOSOPHY

Undergraduate systems and AI engineer at Amrita Vishwa Vidyapeetham with an established track record across **high-performance distributed runtimes**, **agentic MCP protocols**, and **physics-informed machine learning**. Pioneer of the “**Extreme Economic Engineering**” paradigm: combining mathematical first principles and algorithmic efficiency to execute complex neural inference, vector search, and interactive streaming on low-cost host hardware without cloud lock-in. Disciplined NCC cadet, 3-time Services Selection Board (SSB) attendee, and certified in Indian Sign Language (ISL).

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
 - **Relevant Coursework:** Data Structures & Algorithms, Operating Systems Concepts, Computer Networks, Linear Algebra, Probability & Stochastic Processes, Object-Oriented Software Design, Database Management Systems.
 - **Core Focus:** High-Throughput I/O Runtimes, Local Vector Indexing, Physics-Informed ML, Autonomous Agent Protocols.

TECHNICAL SKILLS

- **Programming Languages:** Python, Rust, TypeScript, JavaScript, C/C++17, Go, Dart (Flutter), SQL, POSIX Shell/Bash.
- **Systems, Networking & Runtimes:** Actix-Web, Tokio async, FastAPI, Node.js, WebSockets, PTY orchestration, Linux eBPF, Docker, Linux IPC.
- **Machine Learning, DSP & Agent Protocols:** PyTorch, Scikit-Learn, FastMCP, Model Context Protocol (MCP), Gemini API, Fast Fourier Transform (FFT).
- **Frontend, Mobile & GUI Systems:** React 18, Vite, React Native, Flet (Python GUI), Three.js / React Three Fiber, WebGL, Reanimated, Streamlit.
- **Cloud, Security & Databases:** Google Cloud Run, Firebase Auth (OIDC), Cloud Firestore, PostgreSQL, SQLite Local-RAG, Redis, Linux PAM.
- **Accessibility & Human Languages:** Certified in Indian Sign Language (ISL) fundamentals for non-verbal human-assistive interaction; English, Telugu, Hindi.

LEADERSHIP, COMPETITIONS & HONORS

- **IIT Madras Smart India Challenge – 2nd Runner Up & Youngest Panelist** *IIT Madras*
 - Awarded **2nd Runner Up** at IIT Madras Smart India Challenge for developing high-efficiency distributed and cyber-physical architectures.
 - Invited and served as the **Youngest Panelist** at the summit, presenting on low-overhead systems, edge diagnostics, and sovereign AI computation.
 - Demonstrated **Dust-Vision** (computer vision optical particulate tracking on GitHub) and **Dust-Sim** (real-time physics dispersion kinematics live on the web).
- **High School Autonomous Robotics AI Hackathon – 1st Place Winner** *11th Standard*
 - Clinched **1st Place** in an internal school AI hackathon by developing an autonomous self-parking robot.
 - Solved the core line-following “snaking” oscillation issue through a responsive hardware sensor array and damping algorithms.
 - Orchestrated global floor plan coordinate navigation utilizing the **A* pathfinding algorithm** for deterministic slot assignment and parking.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected technical delegate at the **GARC Chennai Meet**, engaging with automotive engineers on telemetry standards and diagnostics.
 - Participated in sessions on real-time vehicle telemetry pipelines, edge computing diagnostics, and sensor anomaly isolation.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - Trained cadet in the **National Cadet Corps (NCC)**, instilling strong leadership, tactical problem-solving, and disciplined execution.
 - Recommended for and attended the **Services Selection Board (SSB) 3 times**, demonstrating high resilience and situational decision-making.

- **Ideation GOAT – Autonomous Semantic Multi-Domain Architect Agent | Public** *Link : Release v1.0.0*
 - Architected an enterprise-grade **Model Context Protocol (MCP)** server exposing **24 autonomous diagnostic tools** across codebases, academic literature, and UI/UX design.
 - Leveraged **Gemini 3.1 Flash Lite** with custom Zod-based auto-healing middleware to dynamically self-correct malformed LLM tool arguments at runtime.
 - Engineered multi-domain semantic synthesis pipelines enabling autonomous coding agents to audit supply-chain security, validate API compatibility, and scaffold full-stack architectures.
- **ytstr – High-Performance Psychoacoustic Audio Engine | Public** *Link : Release v2.0.0*
 - Developed a terminal-native streaming runtime powered by a psychoacoustic **DecisionEngine** executing real-time spectral analysis.
 - Implemented 8 broadcast-grade transition algorithms (Bass Swap, Frequency Melt, Tape Stop) and built an active backpressure memory pool.
 - Slashed runtime RAM footprint by **70%** (< 260MB peak) during uninterrupted multi-hour streaming.
- **EVolvAI – Generative AI & Physics-Constrained Grid Optimization | Public** *Link : Website | Release v1.0.0*
 - Architected a scalable cyber-physical demand framework in PyTorch coupling a **Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm (GA).
 - Generated thousands of non-linear spatial charging scenarios under distribution grid bus limits and severe climate stress anomalies, achieving $R^2 \approx 0.5$ predictive accuracy on sparse field data.
 - Engineered an interactive Streamlit simulation console enabling municipal engineers to audit transformer stress curves.
- **TurnBack (Activity-Mapper) – Local-First GPS Endurance Tracker | Public** *Link : Release v1.3.0*
 - Built an ultra-lightweight, 100% serverless, battery-optimized mobile GPS tracking app featuring a proprietary **54% Turn-Back Engine**.
 - Mitigated out-and-back route fatigue asymmetry by alerting users when exactly 54% of their time budget remains with distinct audio/haptic cues.
 - Engineered stationary stop detection and **Ramer-Douglas-Peucker (RDP)** trajectory decimation to guarantee 60 FPS rendering under < 5MB RAM overhead on low-end 2GB Android devices.
- **SpecRAG – Zero-Hallucination Firmware Auditing Gateway | Public** *Link : Release v0.1.0*
 - Engineered an enterprise developer tool bridging 1,000-page silicon datasheets with IDE source code using the **Model Context Protocol (MCP)**.
 - Performed localized vector retrieval against indexed silicon register banks to catch peripheral pin collisions, clock-tree violations, and timing hazards directly in VS Code before compilation.
- **MotorSafe – Tri-Modal Distributed Telemetry Engine & Digital Twin | Public** *Link : Release v0.01*
 - Constructed a real-time telemetry streaming system ingesting 3 synchronized sensor channels over MQTT.
 - Designed localized unsupervised **Isolation Forest** models and Fast Fourier Transform (FFT) DSP pipelines paired with an interactive 3D WebGL/Three.js digital twin.

- **LoRa-Connect – Point-to-Point Offline Vehicle Telemetry Link | Private** *Hardware Prototype | Private*
 - Developed a subscription-free, offline point-to-point LoRa telemetry system bridging smartphones directly to vehicles without cellular dependence.
 - Designed a custom USB-C RP2040 dongle establishing an encrypted 915MHz LoRa link to an onboard vehicle ESP32 node integrated into the CAN bus.
 - Achieved sub-mA standby current and bypassed mobile operating system hardware restrictions via USB CDC-NCM protocol encapsulation.
- **Microintegrated Flet Console – Embedded Low-Overhead GUI | Interdisciplinary** *Link : Website | Repository*
 - Designed an ultra-lightweight, cross-platform cyber-physical user interface combining microintegrated **Flet (Python + Flutter)** with embedded microcontrollers.
 - Engineered asynchronous event loops connecting serial UART/USB streams from embedded devices directly into real-time reactive widgets.
 - Delivered responsive 60 FPS hardware telemetry plotting and parameter calibration dashboards on low-spec host systems with zero web browser overhead.
- **Lunel-Gravity – Cloud Development Platform & Remote Mobile IDE | Public** *Link : Website | Release v0.1.0*
 - Architected an AI-native cloud development workspace enabling full-stack coding and terminal execution directly on resource-constrained mobile hardware.
 - Engineered an asynchronous PTY orchestration runtime streaming bi-directional shell sessions over WebSockets with sub-millisecond input latency.
- **LegalEase – Privacy-First Local-AI Heuristic Legal Navigator | Public** *Link : Repository*
 - Constructed an offline, sovereign legal navigation assistant running Cython-accelerated heuristic retrieval over local SQLite indices.
 - Integrated local LLMs to parse complex statutory clauses and provide structured legal explanations without transmitting confidential data to external clouds.
- **Web2API – High-Throughput Headless Reverse Proxy Gateway | Public** *Link : Repository*
 - Developed an asynchronous REST API bridge using FastAPI and Playwright to automate programmatic interaction with conversational AI interfaces.
 - Implemented intelligent token bucket rate-limiting and browser context pooling to eliminate cold starts and maximize request throughput.
- **errands-op (op-list) – Algorithmic Errand Routing Engine & VRPTW Solver | Public** *Link : Repository*
 - Built an offline-first algorithmic routing engine in **Flutter/Dart** solving multi-stop Vehicle Routing Problems with Time Windows (VRPTW) and bi-directional Home Assistant WebSocket synchronization.
 - Engineered 2D Kalman GPS filtering with dual-ring hysteresis geofencing (40m/80m) and learned in-store layout walk orders using Feedback Arc Set (FAS) cycle elimination and Kahn's topological sort.
- **SAFE-Inventory – Low-Power BLE Warehouse Asset Tracking Suite | Public** *Link : Website | Release v1.4.0*
 - Engineered a resilient warehouse asset management system featuring single-product batch scanning and set-and-forget Bluetooth LE hardware pairing.
 - Optimized transceiver duty-cycling on mobile and embedded scanners to provide multi-shift operational battery endurance.

CERTIFICATIONS & TECHNICAL WORKSHOPS

- **Indian Sign Language (ISL) Certification** *Certified*
 - Earned formal certification in Indian Sign Language (ISL) fundamentals, enabling inclusive, non-verbal communication and human-assistive interface design.
- **Hardware Prototyping Workshop & Battery Spot Welder Fabrication** *Practical Engineering*
 - Designed and constructed a high-current capacitive discharge battery spot welder for precise nickel strip bonding and custom lithium-ion battery pack assembly.